UNIVERSITI TEKNOLOGI MALAYSIA · FACULTY OF COMPUTING

GreenStep

PR3 FINAL DEMO · PR4 PRESENTATION

Small daily steps. Measurable climate impact.

Personal Carbon Footprint & Eco-Lifestyle Tracker — a cross-platform Vue 3 + PHP Slim 4 system

Vue 3 | Pinia | PHP Slim 4 | MySQL + PDO | JWT Auth | Capacitor

PROJECT	GROUP	LECTURER	FINAL DEMO
P3 · GreenStep	RM2Z · Section 02	Dr. Hishammuddin B. Asmuni	02 July 2026

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TEAM PROFILE & ROLES

Four leads, one integrated build



Zeng Liting

Frontend Lead · A24CS4073

- Vue 3 SPA scaffold, Router & Pinia
- 7 screen-by-screen views, mobile-first CSS
- Connected every page to the REST API



Mohammed Mohsen Alsakkaf

Backend & API Lead · A23CS4026

- PHP Slim 4 API · 7 controllers · 20 routes
- JWT middleware, CO2 + badge engines
- JSON envelopes, error handling, API docs



Rawan Mohamed Salih Magzob

Database & Security Lead · A24CS4066

- Authoritative MySQL schema & ER model
- Hardened PDO layer, prepared statements
- JWT/role policy, SECURITY threat model



Zineb Ennabili

DevOps, Mobile & Integration · A24CS4057

- Deployment lead · Render / Netlify / Cloud SQL
- Capacitor Android wrap, env config
- Integration testing, bug-fix sweep, demo orchestration

GitHub Evidence

github.com/ennabilizineb/GreenStep

106 COMMITS · 4 CONTRIBUTORS · CLEAN HISTORY



Branching: work merged to `main`; release deliverables isolated on the **PR3** branch.

Existing carbon apps miss the middle

A substantial share of global emissions traces back to everyday choices in **transport, diet and energy**. Yet people rarely get immediate, quantifiable feedback so habits never form. This platform came to make a change in people's daily small habits encouraging them with challenges & badges.

Oversimplification

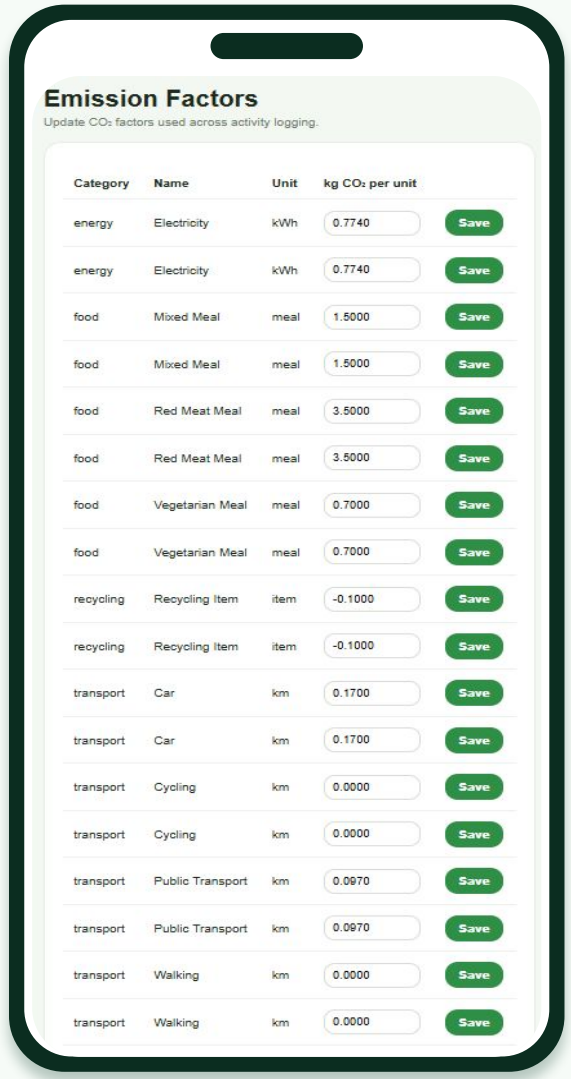
Failure mode 01

Static "green tips" ("turn off the lights"). No tracking, no calculation, no feedback loop — nothing to motivate a habit.

Over-Complication

Failure mode 02

Scientific calculators demand full life-cycle assessments — utility bills, exact distances, receipts. Cognitive overload → burnout → abandonment.



"Individuals lack immediate, quantifiable feedback on the carbon implications of their daily actions."

THE GAP

No app combines **low-friction daily logging** with **empirical, scientifically-grounded CO_2 calculations.**

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OUR SOLUTION · GREENSTEP

Turn accountability into a 60-second daily habit

Log transport, meals, electricity and recycling with single-tap inputs. A PHP Slim 4 engine computes the footprint server-side against database-driven DEFRA / IPCC emission factors, then drives streaks, badges and community challenges.

OBJECTIVE 1

Multi-Role Secure Access

RBAC for User · Leader · Admin, all behind stateless JWT + bcrypt.

OBJECTIVE 2

Server-Side Calc Engine

Slim 4 REST API computes CO_2 from DB-driven factors via PDO.

OBJECTIVE 3

Low-Friction Cross-Platform

Vue 3 SPA, mobile-first, wrapped as an Android APK with Capacitor.

OBJECTIVE 4

Interactive Visualization

Weekly & monthly trends rendered from live JSON dashboards.

OBJECTIVE 5

Gamification & Retention

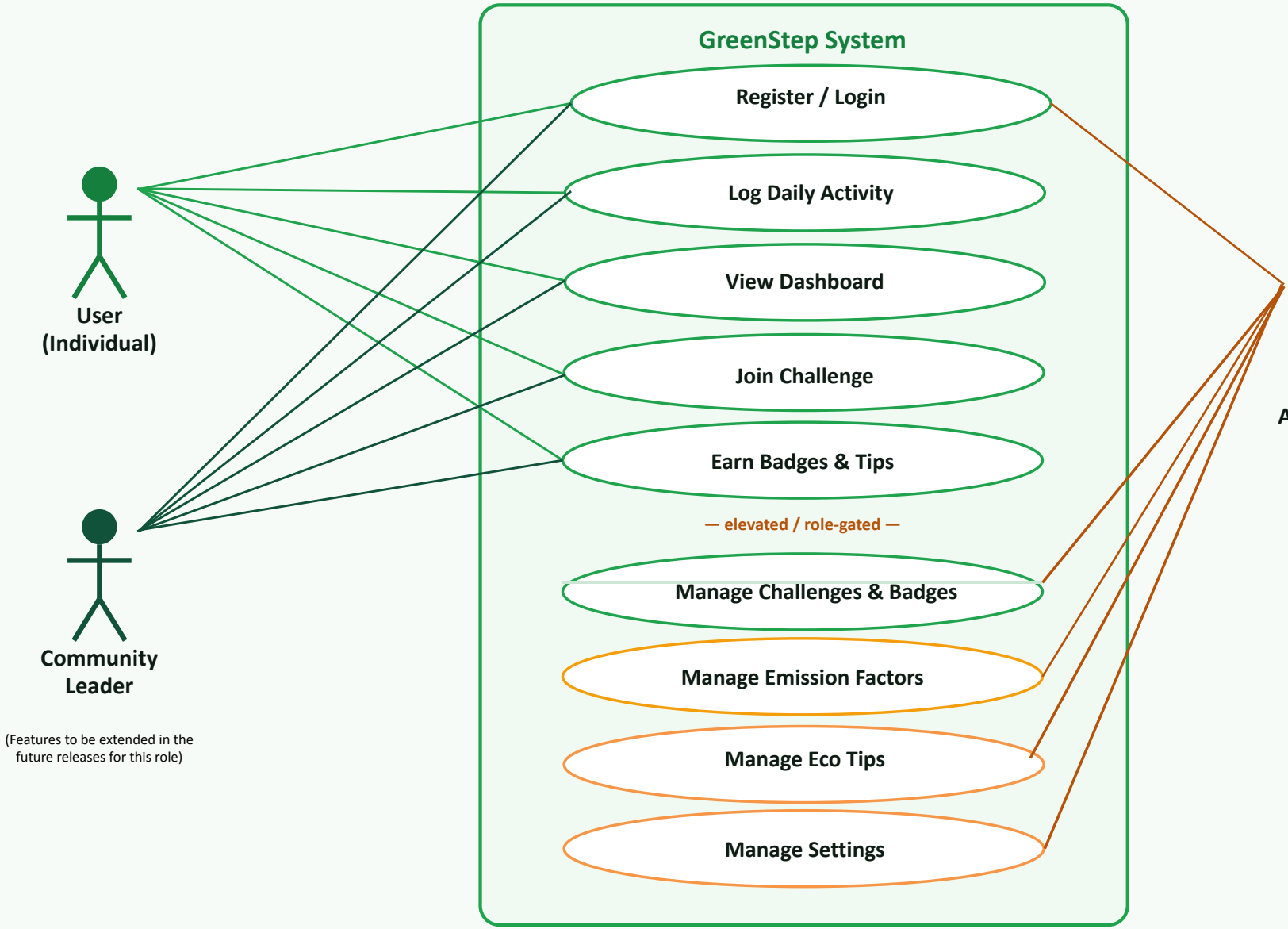
Automated streaks, JSON-criteria badges, collective challenges.

Scope delivered: Auth · Daily Log (full CRUD) · Dashboard · Challenges (CRUD) · Badges · Daily Tips · Admin factor & tip control — 3 roles, 10 tables, 20 endpoints.

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TARGET USERS & USE CASES

UML use-case diagram — three actors, one system



(Features to be extended in the future releases for this role)

As a user/leader, I want to log daily activities easily, so that I can track my carbon footprint.
As a user/leader, I want to view my dashboard, so that I can understand my daily and weekly CO_2 trend.
As a user/leader, I want to earn badges and join challenges, so that I feel motivated to build eco-friendly habits.

As an administrator, I want to manage eco-tips and emission factors, so that the system remains accurate and useful.

These user stories are translated into the user cases shown in the case diagram.

ROLE SCOPE (inherits downward)

U — User
Logs activities, views dashboard, earns badges, joins challenges.

L — Leader
Same as User role currently. For future sprint we will try to implement: all User rights + create / update / delete community challenges

A — Admin
All rights + curate tips & alter the scientific emission factors.

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TECHNOLOGY STACK

A decoupled, full-pipeline cross-platform system

PRESENTATION
Vue 3 SPA · Composition API

Vue Router 5 | Pinia 3 | Vite 8

MOBILE
Single codebase → native Android

Capacitor | Android APK | iOS-ready

APPLICATION / API
PHP 8.1 · Slim 4 RESTful micro-framework

9 Controllers | JWT Middleware | PDO · JSON

DATA
MySQL 8 relational store

11 tables | PDO prepared | FK cascade

DEVOPS / DELIVERY
Local Laragon → cloud

GitHub | Render · Netlify | Cloud SQL

WHY THIS STACK

Decoupled
Frontend is pure presentation; the API is a headless, stateless engine.

Authoritative
All CO_2 maths run server-side — clients never compute footprints.

Portable
One Vue codebase ships to web and Android via Capacitor.

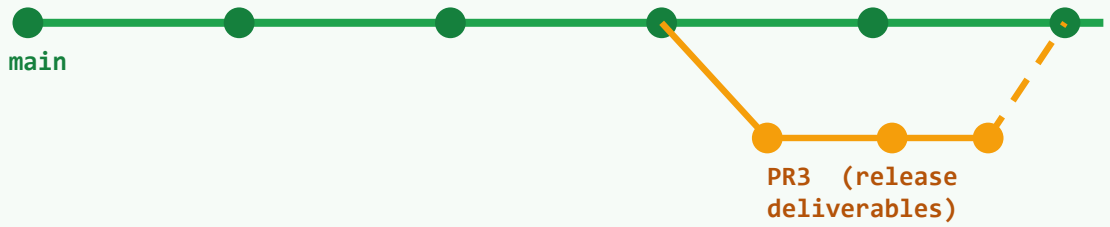
Free-tier
Render + Netlify + Cloud SQL = a public URL at zero hosting cost.

One repository, four contributors, clean history

github.com/ennabilizineb/GreenStep

Monorepo — Backend/ (Slim 4) + Frontend/ (Vue 3) + SQL + Capacitor + apk, README doc.

BRANCH STRATEGY



All members commit directly, merge into **`main`**; PR3 slides, README doc are staged on a dedicated **PR3** branch.

CONTRIBUTION EVEDENCE

106 commits by verified author e-mail



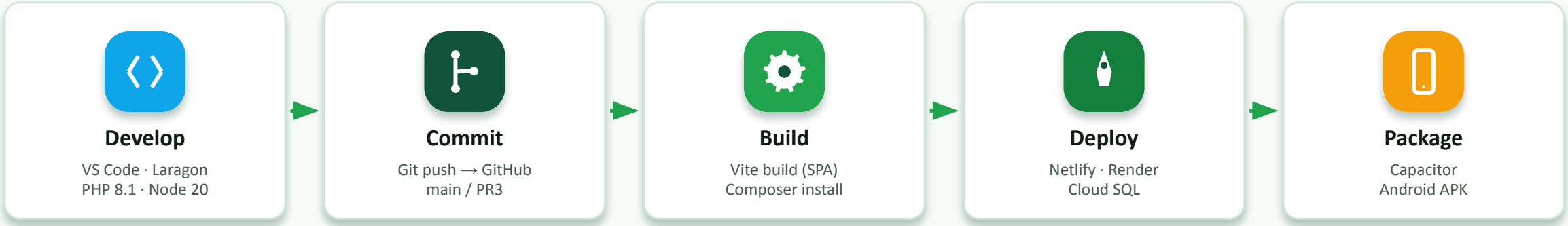
REPRESENTATIVE COMMITS

- **Backend:** Implement badge evaluation & collective challenge progress
- **DevOps:** logout() missing! · Redirecting issues fixed! · Daily Log bug-fix
- **Frontend:** Connect dashboard & challenges pages to backend API
- **Database:** update database schema and seed data · SECURITY.md threat model

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DEVOPS · CI/CD & DEPLOYMENT

From a Laragon laptop to a public cloud URL



Deployment Architecture

Frontend — Netlify
Static Vue 3 build, auto-rebuild on push

Backend API — Render
PHP Slim 4 service over HTTPS

Database — Google Cloud SQL
Managed MySQL 8 via PDO

Mobile — Capacitor via Android Studio
Vue build wrapped as Android APK

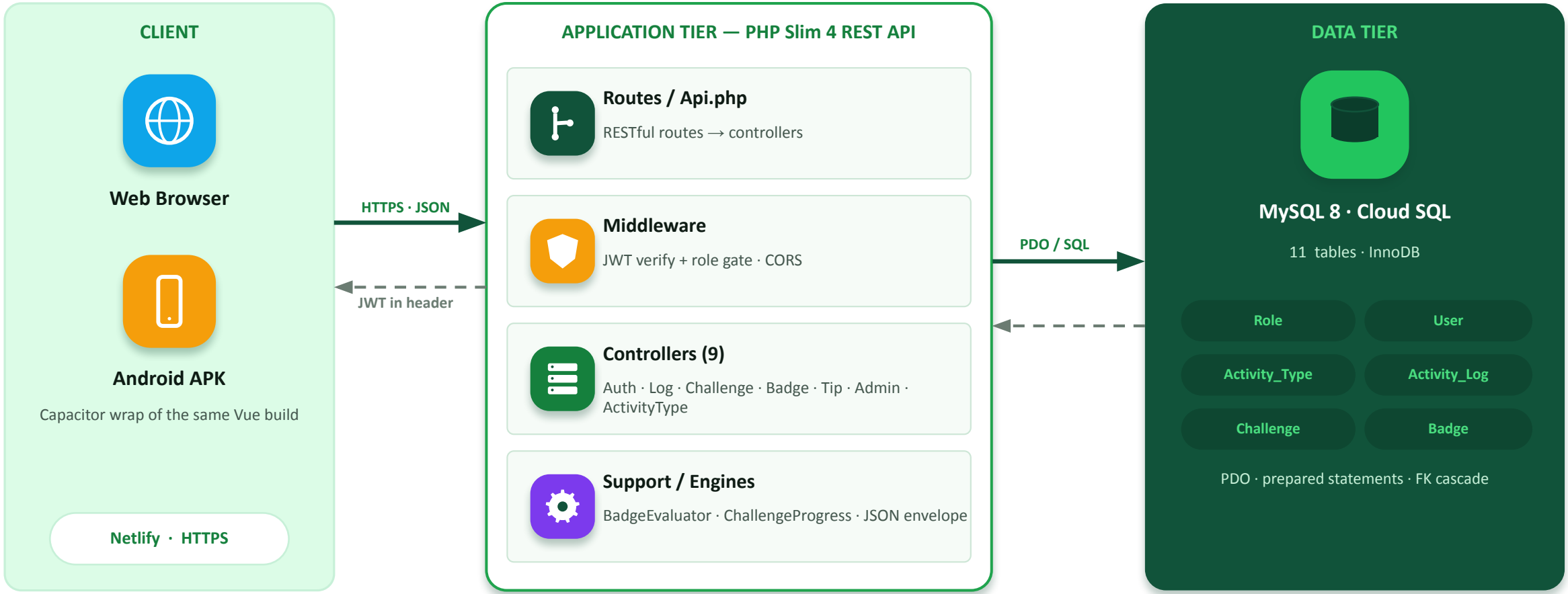
TOOLCHAIN & STATUS

Laragon · Composer · npm / Vite · Postman · HeidiSQL · GitHub · VS Code · Capacitor

CI signal: Vue lint (oxlint + ESLint) and `composer validate` gate every build; Postman collection verifies all 20 endpoints before the demo.

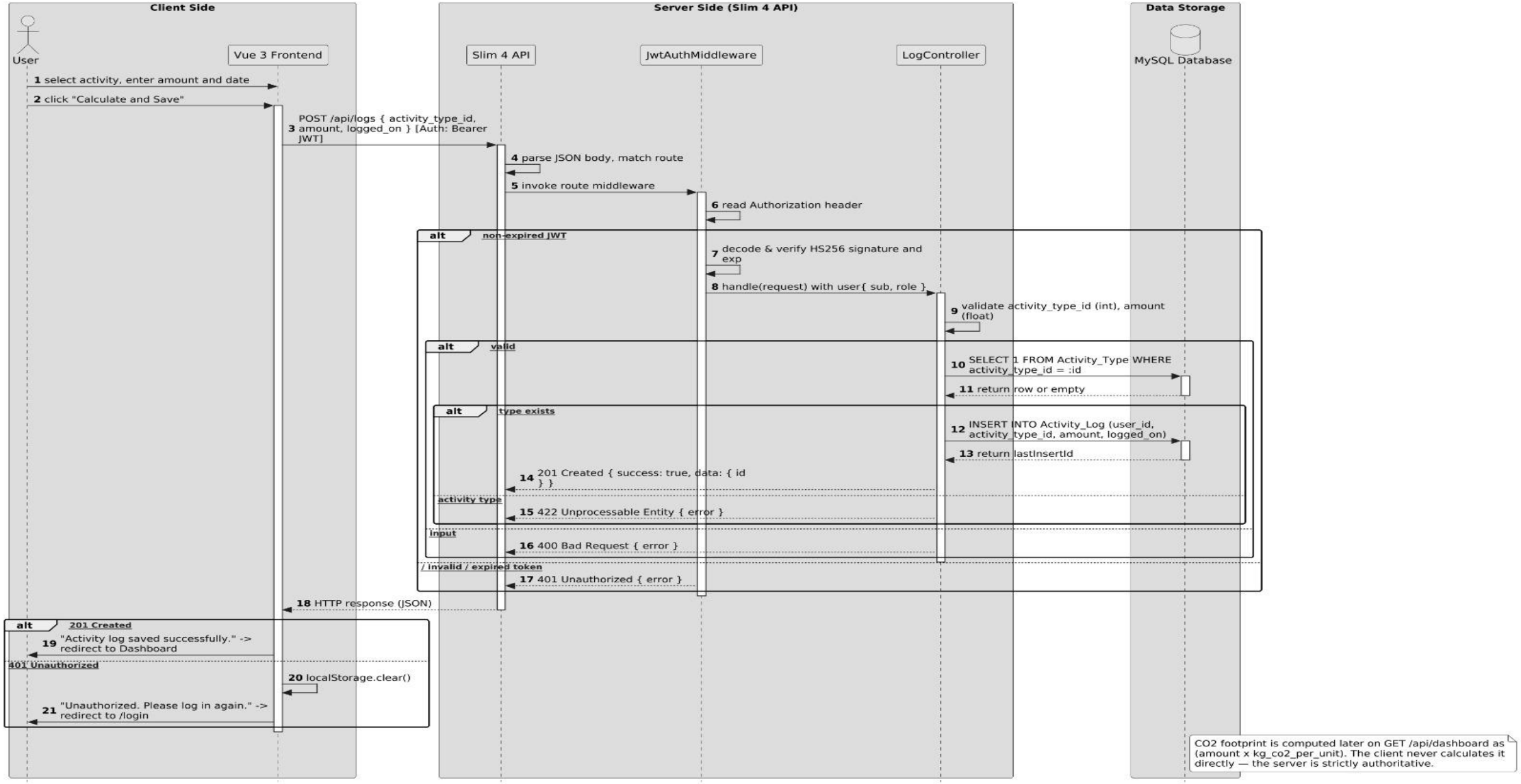
SYSTEM ARCHITECTURE

Three decoupled tiers communicating over REST + JWT



SEQUENCE DIAGRAM (UML Notation)

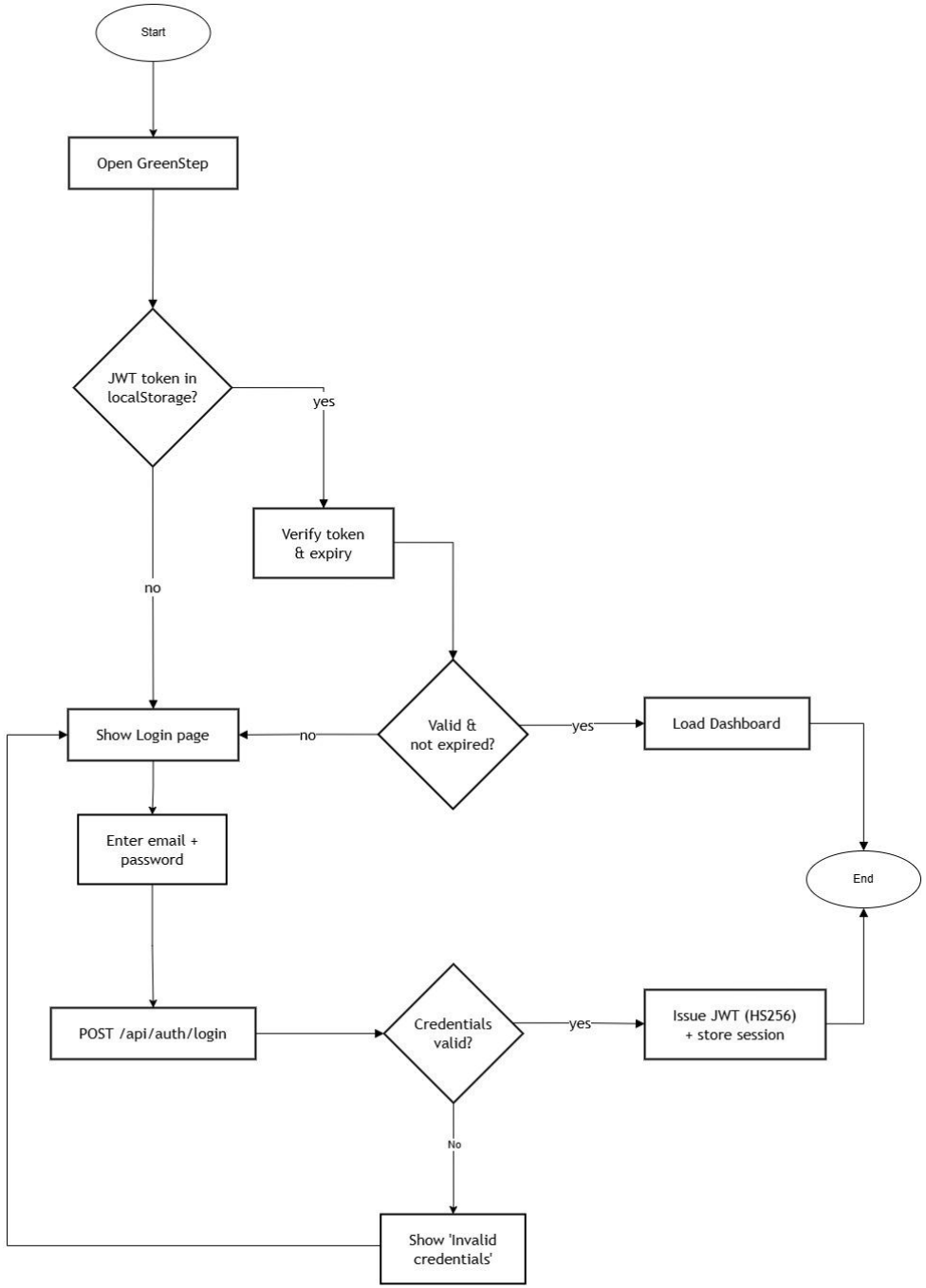
Daily Logs Calculations — logging a daily activity, end to end



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ACTIVITY DIAGRAM (UML Notation)

User authentication & session flow



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ACTIVITY DIAGRAM (UML Notation)

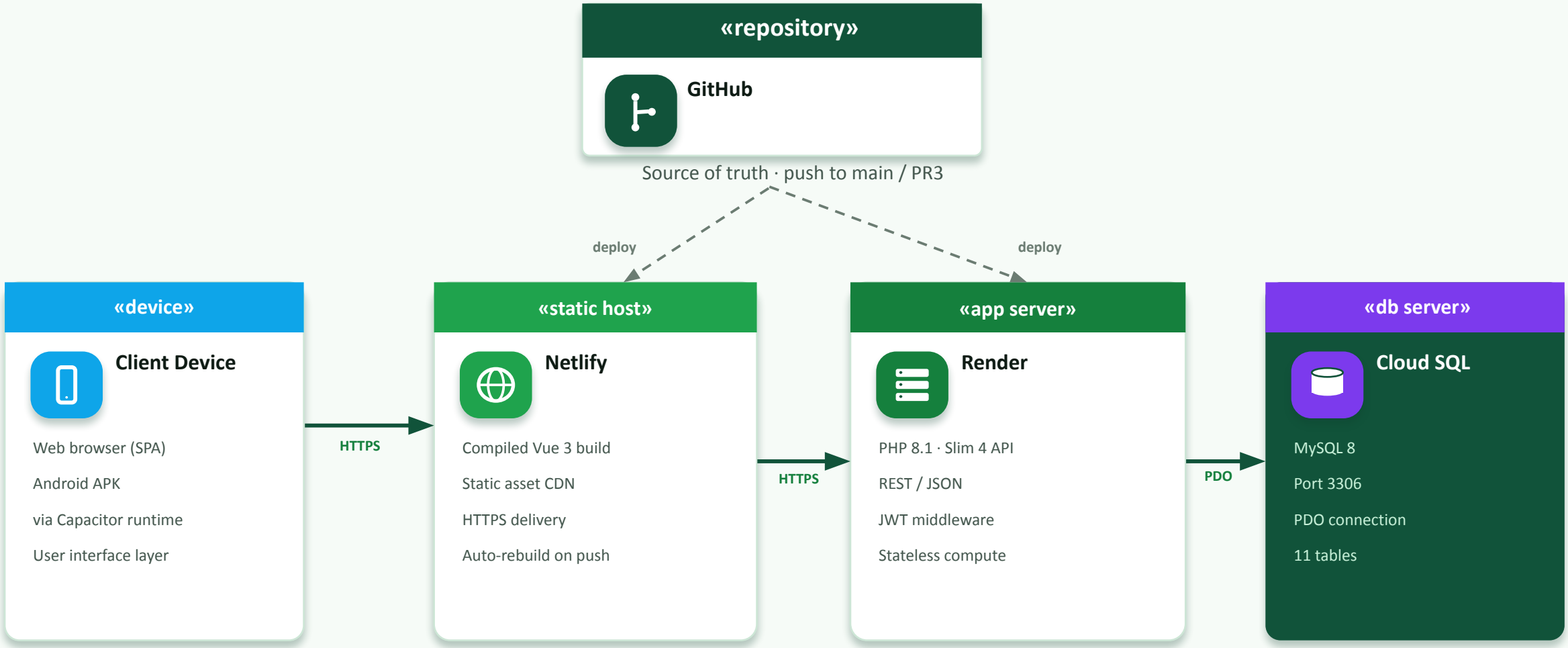
Daily Activity Log Flow



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DEPLOYMENT DIAGRAM

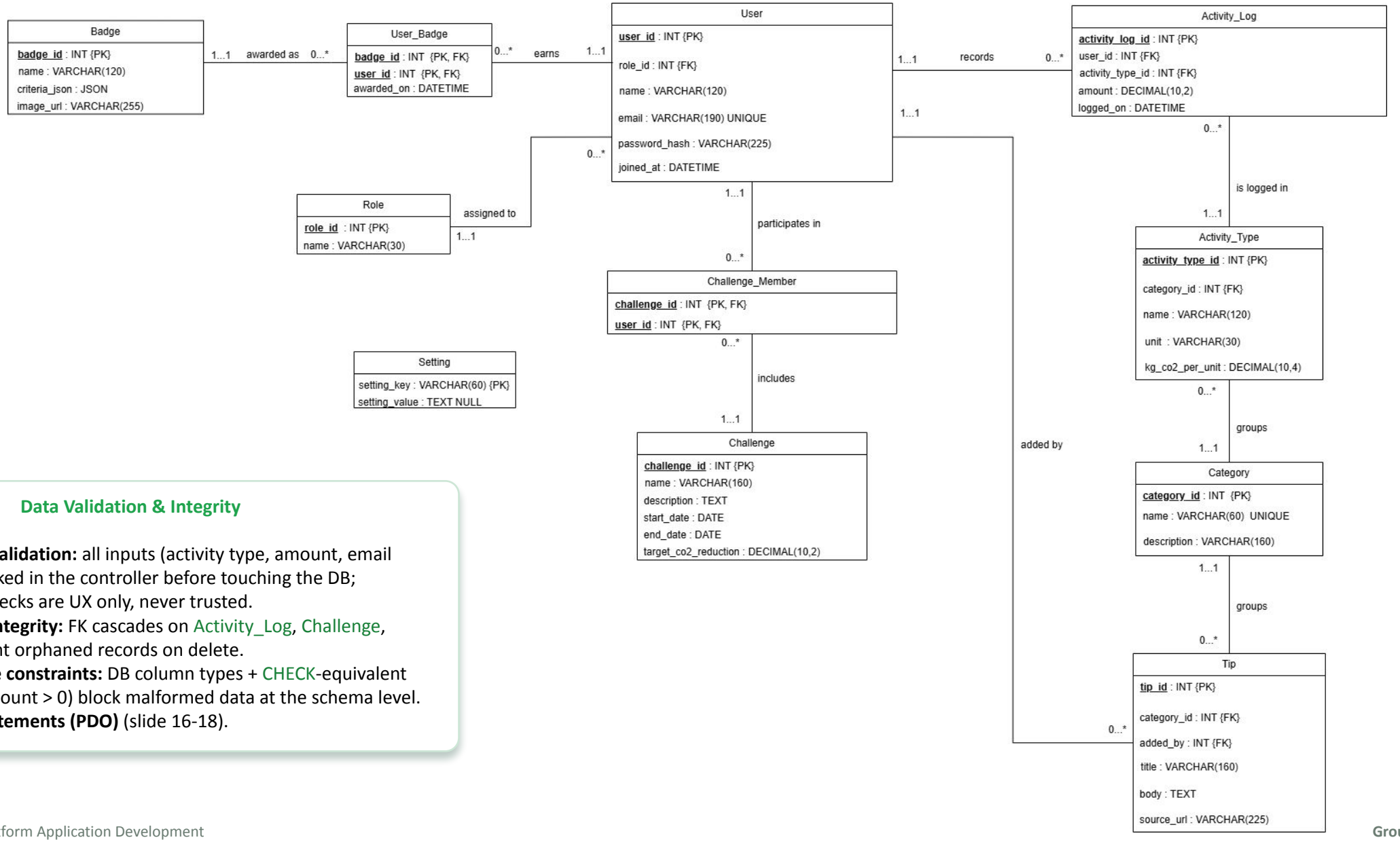
Nodes, hosting environments & network protocols



Free-tier public deployment: the browser/APK loads the Vue SPA from Netlify, which calls the Slim API on Render over HTTPS; the API reaches MySQL on Cloud SQL via PDO. GitHub drives every deploy.

DATABASE · ENTITY-RELATIONSHIP DIAGRAM

11 normalized tables, full referential integrity



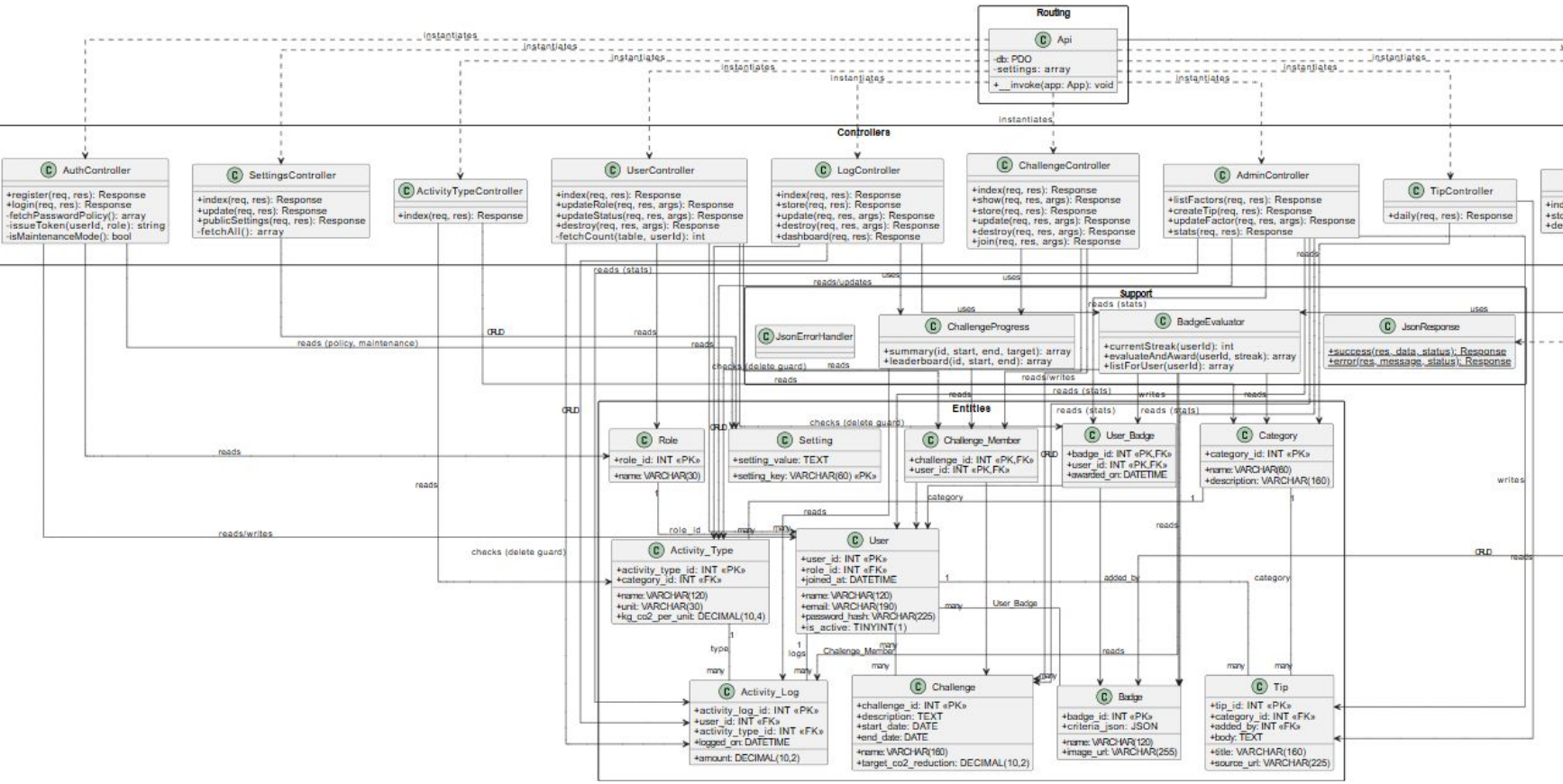
Data Validation & Integrity

- **Server-side validation:** all inputs (activity type, amount, email format) checked in the controller before touching the DB; client-side checks are UX only, never trusted.
- **Referential integrity:** FK cascades on Activity_Log, Challenge, Badge prevent orphaned records on delete.
- **Type & range constraints:** DB column types + CHECK-equivalent logic (e.g. amount > 0) block malformed data at the schema level.
- **Prepared statements (PDO)** (slide 16-18).

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Class Diagram

Controller, Entity & Service Structure



The class diagram represents the main backend structure of GreenStep.
Controllers process API requests, entity classes represent the main data objects, and service classes handle shared logic such as authentication, database connection, badge evaluation, and challenge progress.
It also shows the relationships between users, activity logs, challenges, badges, tips, and activity types.

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BACKEND ENGINE · REST API & SECURITY

33 endpoints · 9 controllers · stateless JWT

API SURFACE — grouped by access level

3 **Public**

`/health · /auth/register · /auth/login`

11 **Authenticated User**

`logs CRUD · dashboard · activity-types · tips · badges · challenges · join`

3 **Community Leader**

`challenges: create · update · delete`

3 **Administrator**

`tips: create · factors: list & update`

Inheritance: Leader = User ⊇ Public · Admin ⊇ all. Wrong role → **403**; missing/expired token → **401**.

Database.php

```php
// hardened PDO — structural SQLi defence
new PDO($dsn,$user,$pass, [
  PDO::ATTR_ERRMODE =>
      PDO::ERRMODE_EXCEPTION,
  PDO::ATTR_EMULATE_PREPARES => false,
]);  // params can never become SQL
```

AuthController.php — issueToken()

```php
$payload = [
  'sub'=>$id, 'role'=>$role,
  'iat'=>$now, 'exp'=>$now+$ttl ];
JWT::encode($payload,$secret,'HS256');
```

- bcrypt password_hash()
- time-safe verify
- HS256 JWT · 1h TTL
- user-enumeration safe
- FK cascade integrity
- uniform JSON envelopes

CO₂ ENGINE & GAMIFICATION

Authoritative server-side maths, credible factors

$$CO_2\ (kg) = amount \times kg_co2_per_unit$$

Activity	Unit	kg CO_2
Car	km	0.1700
Public Transport	km	0.0970
Electricity	kWh	0.7740
Red-Meat Meal	meal	3.5000
Mixed Meal	meal	1.5000
Vegetarian Meal	meal	0.7000
Recycling	item	−0.1000

Admins update them via API, never in code. Negative = net carbon saving.

All references agreed on these values. Some of them:

https://www.climatiq.io/data/emission-factor/7232e035-6038-41c5-855f-c8b21898a39d
https://ourworldindata.org/grapher/carbon-footprint-travel-mode
https://myenergystats.st.gov.my/documents/d/guest/grid-emission-factor-gef-in-malaysia
https://www.epa.gov/waste-reduction-model/documentation-waste-reduction-model
https://ourworldindata.org/grapher/ghg-per-kg-poore
https://ourworldindata.org/faqs Ou

Badges

5 badges via JSON criteria — total_logs · streak_days · category_logs — auto-awarded (INSERT IGNORE).

Streaks

Consecutive-day algorithm walks back from today over distinct log dates.

Challenges

Collective baseline-vs-during window: Σ saved_kg vs target, ranked leaderboard.

gamification rule

```
// Badge.criteria_json — evaluated server-side
{ "type": "streak_days", "threshold": 7 }
→ 'Green Streak' unlocks at 7 consecutive days
```

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LIVE DEMO · STEP 3

Challenges, badges & the admin control panel

Eco-Commuter Sprint

Public transit / cycling / walking for 7 days.

58% 3 days left

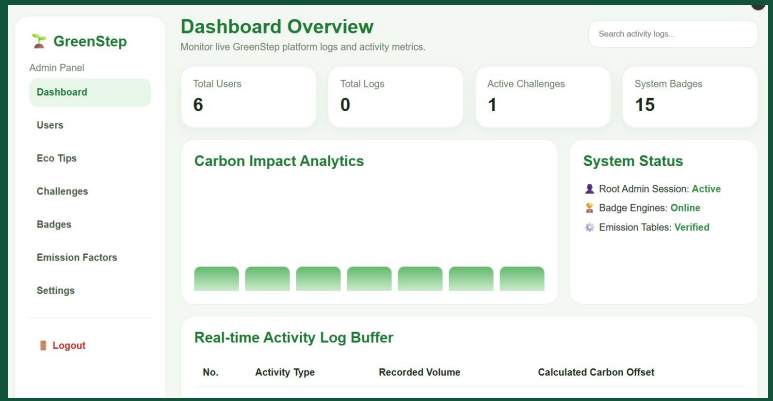
15 participants · target 15.5 kg saved

View Details

Badges — earned & locked

- Eco Beginner
- Green Streak
- Recycling Pro
- Carbon Cutter
- Eco Warrior

ADMIN CONTROL PANEL



Collective progress

Challenge cards show member_count, days_left and a baseline-vs-during % computed across all members.

Automated rewards

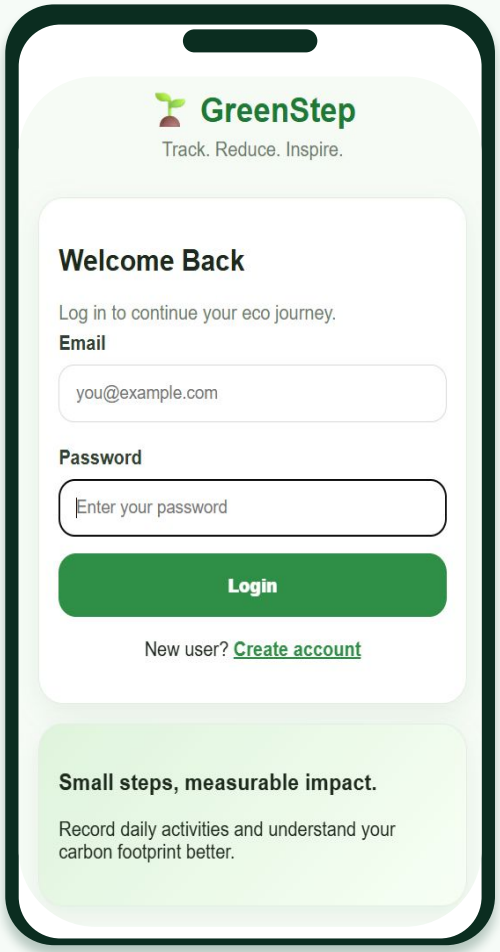
Badges unlock from JSON criteria the moment thresholds are crossed — no manual awarding.

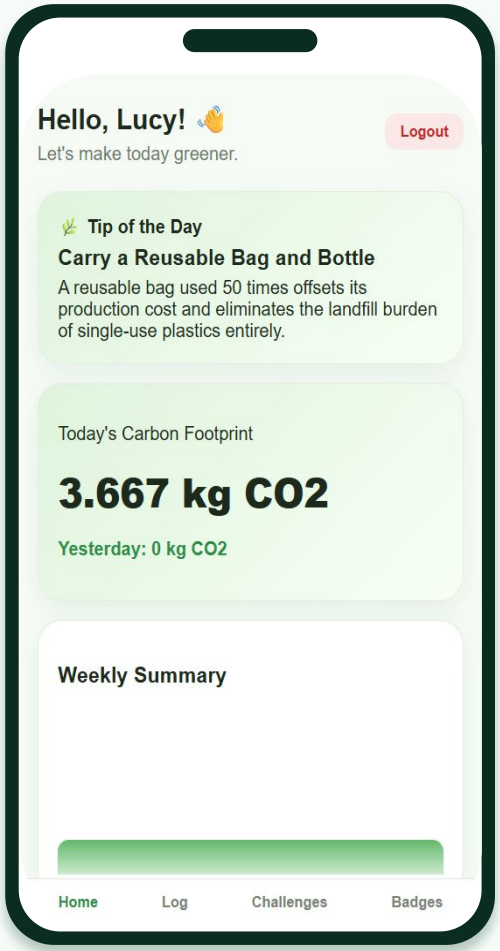
Credible & governed

Admins curate tips and update the very emission factors that power every calculation.

LIVE DEMO

Authentication → personal dashboard





What the panel will see

- Login posts credentials; the API verifies bcrypt + issues a JWT, stored client-side.
- Dashboard calls one endpoint; the server returns today's CO_2, the 7-day trend, streak & badges.
- Every number is computed live from seeded, realistic data — no hardcoded screens.

ENDPOINTS EXERCISED

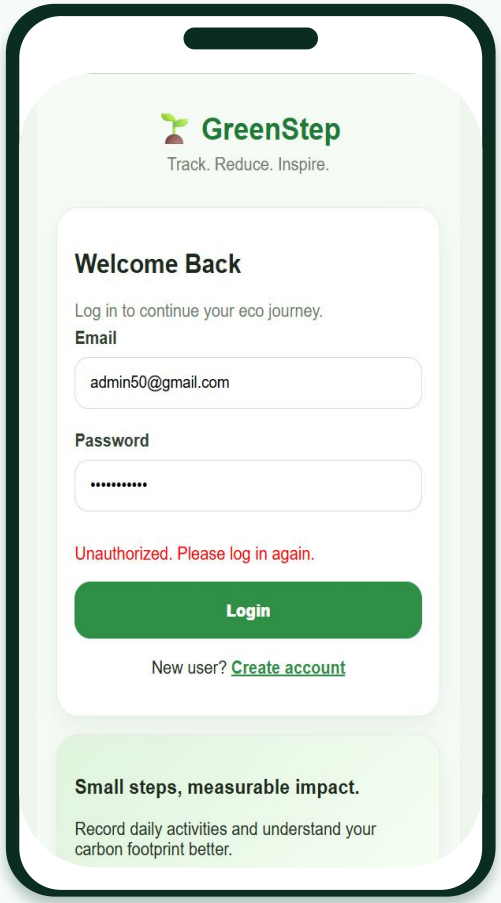
```
POST /api/auth/login   →  200 { token, user }
GET  /api/dashboard    →  200 { today, week, badges }
```

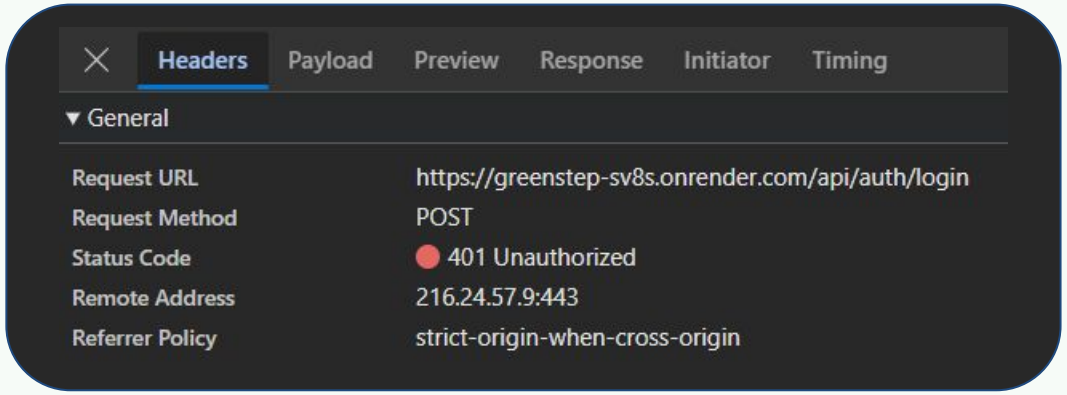
20

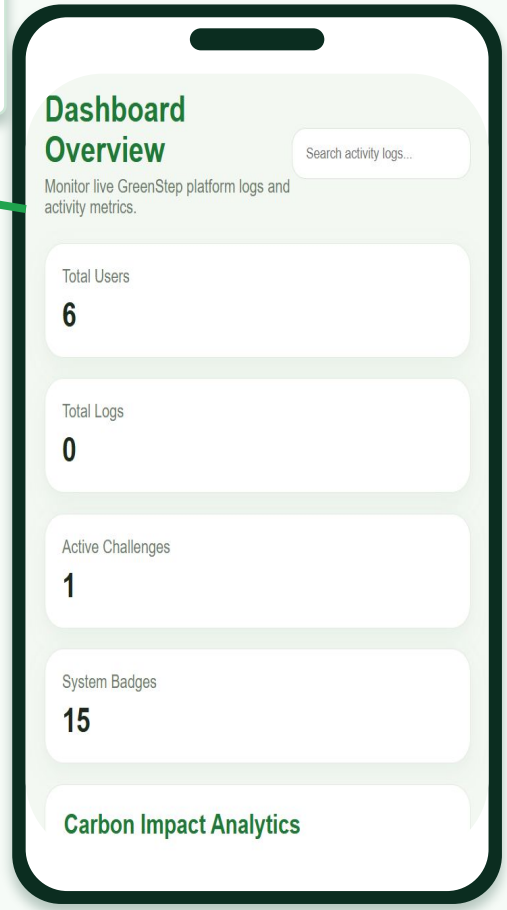
LIVE DEMO

Error Handling & Search Bar

The Search Bar was embedded in Admin view in dashboard due to the huge amount of daily logs!







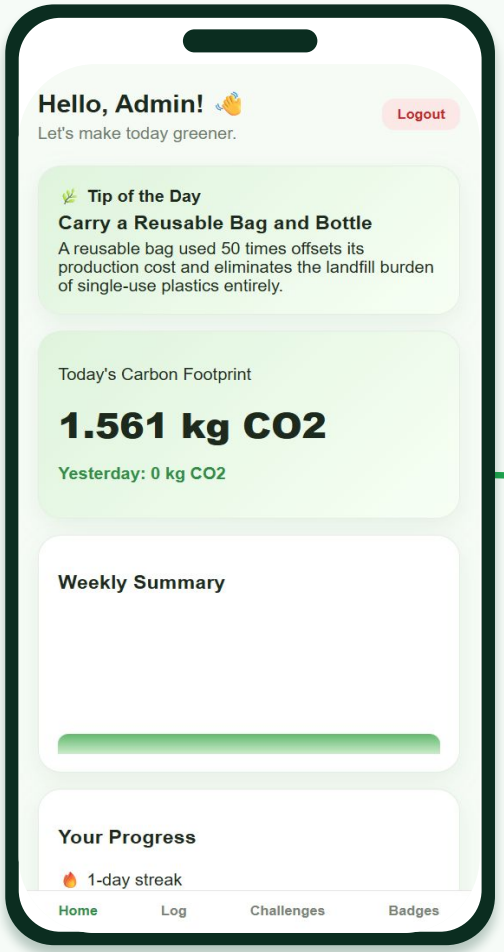
Standardized Exception Handling

- **Slim 4 Error Middleware:** Catches all uncaught exceptions.
- **Standardized JSON:** Returns { status, message, data }.
- **HTTP Status Codes:** like 401 (Auth), 404 (Route), 422 (Validation), 500 (Fatal).

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LIVE DEMO · STEP 2

Log an activity → footprint updates instantly



Server response

```
POST /api/logs
201 Created
{ "id": 142 }
```

Bus 15 km × 0.044

= 0.66 kg CO_2

Dashboard re-fetches → today updates

Full CRUD, real calculation

- Activity types populate the form from the DB (GET /api/activity-types).
- Saving validates input, confirms the type exists, then inserts the log.
- The dashboard recomputes the footprint server-side on next fetch.

ENDPOINTS EXERCISED

```
GET  /api/activity-types
POST /api/logs   →  201 { id }
GET  /api/dashboard (recompute)
```

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PROJECT ACCESS & TESTING

Repositories, live access & test credentials

For more functionality testing, run the live frontend link in your browser and use the following credentials!

GitHub Repository

Monorepo · main branch · deliverables on PR3

Full Project	`github.com/ennabilizineb/GreenStep`
Frontend	`github.com/ennabilizineb/GreenStep/tree/main/Frontend`
Backend	`github.com/ennabilizineb/GreenStep/tree/main/Backend`

Live System Access

Public Vue 3 frontend (Netlify)

Live Frontend	*https://greenstepproject.netlify.app/*
Web App on Android Studio Demo	*https://youtu.be/hyS4yHoPnnA*

Test Credentials

All 3 roles · login identifier = email

U — User role 1
email: `member@greenstep.com`
pass: `password`

L — Leader role 2
email: `leader@greenstep.com`
pass: `password`

A — Admin role 3
email: `admin50@greenstep.com`
pass: `Password123`

Note: accounts are provisioned by **`seed.sql`** — run it on the deployed database before testing. Passwords shown are the seeded demo values.

ENTREPRENEURIAL PITCH

From course project to commercial product · CLO3

CLO3

REVENUE MODEL 1

B2B Corporate SaaS

- Companies & universities chase ESG targets and staff engagement.
- Paid tier: private, closed-loop sustainability challenges for employees.
- Corporate dashboard with anonymized carbon analytics for reporting.

REVENUE MODEL 2

Affiliate Brand Partnerships

- Consumer app stays free; streaks & wins earn redeemable points.
- Points ⟶ discount codes from vetted eco-friendly brand partners.
- Revenue from partnership placement fees + affiliate commissions.

WHY USERS WILL TRUST THE NUMBERS

Institutional baselines

Factors sourced from DEFRA UK & IPCC AR6 — not invented.

Decoupled & governed

Factors live in the DB, admin-controlled; ready to pull live open-data registries.

Transparent methodology

Every figure shows its formula; optional camera proof for eco-actions.

CONCLUSION

A complete, integrated cross-platform system

11	33	3	106	4
Tables	Endpoints	Roles	Commits	Leads

WHAT WE BUILT

- Vue 3 + Slim 4 + MySQL + JWT, fully integrated
- Server-side CO_2 engine on DEFRA / IPCC factors
- Gamification: streaks, JSON badges, group challenges
- Android APK via Capacitor + Google SQL Cloud deploy (DB) + Netlify (Frontend) + Render (Backend)

WHAT WE LEARNED

- Cross-platform delivery from one codebase (CLO1)
- Secure REST API design — JWT, bcrypt, PDO (CLO2)
- An entrepreneurial, deployable product mindset (CLO3)
- Real Git collaboration across four contributors

CLO1 · Cross-platform | **CLO2 · Innovative solution** | **CLO3 · Entrepreneurial**

Thank you

Team RM2Z — Questions welcome

github.com/ennabilizineb/GreenStep · SCSM2223 · Section 02 · 2025/2026

Zeng Liting · Mohammed Alsakkaf · Rawan Mohamed Salih · Zineb Ennabili